

Heaven's Light is Our Guide

Rajshahi University of Engineering & Technology

Department of Computer Science & Engineering



LAB REPORT

Course Code : CSE 4110
Course Name : Information System Design & Analysis Sessional
Experiment No. : 2
Experiment Name : Perform Initial Investigation of the Information System

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Chapter 2

Initial Feasibility Study

2.1 Introduction

This chapter presents the initial feasibility study of the PoC-RUET directory application. The chapter examines the practicality of the app as a digital campus directory and communication support system for students, faculty members, officers, and staff of Rajshahi University of Engineering & Technology (RUET). Identify early-stage issues that influence the usefulness, accessibility, trustworthiness, and long-term sustainability of the system in a real university environment.

The purpose of this chapter is to evaluate whether the current concept and feature set of the app are sufficient to support day-to-day institutional communication. By analyzing the existing limitations of the directory from both technical and operational perspectives, this feasibility study highlights the most important challenges that may reduce user adoption, weaken confidence in the system, and limit the app's ability to function as a reliable campus-wide service.

2.2 Initial Feasibility Analysis

The initial feasibility analysis is used to assess whether the PoC-RUET app can realistically operate as a reliable, scalable, and user-friendly university contact directory. This analysis considers the present architecture, expected user behavior, maintenance demands, and institutional requirements to determine whether the application is practical for broader implementation and future improvement.

A preliminary review of the PoC-RUET directory application indicates that the system has significant promise, but its success depends on how effectively a set of core limitations is addressed. The feasibility assessment can be understood through the following dimensions:

1. **Technical Feasibility:** The app must maintain accurate, searchable, and consistently available directory information while performing well under real campus conditions, including weak connectivity, frequent data changes, and growing data volume.
2. **Economic Feasibility:** The system should reduce repetitive manual work, moderation overhead, and administrative dependency so that the cost of maintaining and scaling the platform remains reasonable.
3. **Operational Feasibility:** The platform must support the needs of a diverse user base, including faculty members, students, officers, and lower-level support staff, without introducing barriers to access or use.
4. **Legal and Ethical Feasibility:** Profile information, communication data and uploaded photos must be managed responsibly, taking into account privacy, verification, and institutional policy.
5. **Schedule Feasibility:** Enhancements should be implementable in manageable phases so that the most urgent weaknesses can be addressed first without delaying the overall progress of the application.

2.2.1 Problem 01

Statement of the problem

Data accuracy and stale information within the directory represent a major problem in the PoC-RUET app. This issue occurs when staff details change at the institutional level, but the app database remains static or outdated, ultimately affecting the reliability and credibility of the entire system.

Summary of findings

Users may encounter incorrect phone numbers, outdated designations, inaccurate office locations, or incomplete staff records because administrative changes are not reflected in the app quickly enough. As a result, the directory may not provide reliable contact information at the time it is needed.

Details of findings

In a university environment, personnel information changes frequently due to promotions, transfers between departments, role adjustments, retirements, and new appointments. When the PoC-RUET

app is dependent on manual data entry or delayed batch updates, the system can quickly fall behind the current institutional record. Even a small delay between an official administrative change and its appearance in the app can create confusion for users trying to contact a staff member urgently.

This problem weakens the practical value of the platform because the usefulness of a directory depends primarily on the correctness of its information. If users repeatedly encounter invalid numbers or outdated office assignments, they begin to question the dependability of the application as a whole. Over time, this can force students, staff, and visitors back to manual inquiries, departmental phone calls, or informal social channels, reducing the effectiveness of the app as a centralized digital directory.

Recommendations

Automated synchronization should be implemented between the PoC-RUET database and the central university HR or payroll systems so that official changes can be reflected in the app with minimal delay. In addition, a user-facing “Suggest an Edit” or “Update My Profile” feature should be introduced so that staff members can flag inaccuracies or request profile corrections directly. Together, these two measures would reduce update latency, improve data integrity, and help maintain user trust in the platform.

2.2.2 Problem 02

Statement of the problem

Rigid search functionality and filtering limitations create another important problem within the app. This issue appears in the main search interface, where users often struggle to locate contacts unless they already know the exact name, spelling, or formal designation of the person they are searching for.

Summary of findings

Users often fail to find the intended contact when they misspell a name, search by a partial title, or rely on a responsibility-based query rather than an exact identity match. The current behavior suggests that the search process is too dependent on strict string matching and lacks flexible discovery support.

Details of findings

People with different levels of familiarity with campus personnel. New students, guardians, visitors, and even members of other departments may know only partial information about a person. For instance, a user may search for “CSE Head,” “Dean Engineering,” or a name with a minor spelling error, expecting the system to infer the intended result. If the app returns no meaningful matches in such situations, the search experience becomes frustrating and inefficient.

This limitation reduces the accessibility of the directory because the burden of precision is placed entirely on the user. Instead of helping users find the correct person through approximations, role-based search, or filtered exploration, the app behaves like a narrow database lookup tool. As a result, users who do not know the exact spelling or formal designation may conclude that the desired person is absent from the system, even when the record is actually present.

Recommendations

The app should incorporate fuzzy search logic so that minor spelling errors, phonetic variations, and partial matches can still return relevant results. Advanced filtering options should also be added to allow users to search by research interests, laboratory affiliation, administrative responsibilities, academic departments, or role categories. These improvements would make the search process more inclusive, more intuitive, and better aligned with real user behavior.

2.2.3 Problem 03

Statement of the problem

An architectural limitation causing a lack of offline functionality is another significant concern. Currently, the app depends heavily on a continuous online connection and does not yet provide built-in offline caching, making it unavailable in areas with poor reception.

Summary of findings

The app may fail to load contact data or search results in known dead zones of the campus, such as basement areas, deep laboratory spaces, or locations with weak Wi-Fi or unstable mobile signals. This means users must depend on an active and stable connection at all times.

Details of findings

Campus connectivity is not always consistent, especially in enclosed buildings, remote corners of departments, or during periods of network congestion. In such conditions, a fully cloud-dependent directory becomes unreliable at the exact moments when quick access to information may be most important. A student may need an office room urgently, a visitor may need help finding a staff member, or a user may need immediate access to an emergency contact number. If the app cannot respond without internet access, its real-world effectiveness becomes severely limited.

This issue affects not only convenience but also system resilience. A directory app that becomes unusable during connectivity interruptions cannot fully support a campus environment where users move across buildings and rely on the app in unpredictable conditions. The absence of offline capability therefore reduces both accessibility and service continuity.

Recommendations

Local data caching should be prioritized in future development. The app should download a compressed and encrypted version of essential directory data, including names, phone numbers, and office locations, onto the user's device for offline querying. Whenever the device reconnects to a stable network, the data can be synchronized silently in the background. This approach would ensure that the most critical directory features remain available even in low-connectivity environments.

2.2.4 Problem 04

Statement of the problem

An inclusion gap for lower-level support staff presents a structural problem in the system. It is currently difficult to integrate 3rd and 4th-class employees cleanly into the central digital directory when access depends on the same formal identity and authentication structure used for other groups.

Summary of findings

Lower-tier support staff may be absent from the directory or unable to access the app because the authentication and profile management process appears too rigid for their practical circumstances.

Details of findings

If the PoC-RUET app relies heavily on official `ruet.ac.bd` email-based single sign-on, it may unintentionally exclude important categories of employees such as guards, cleaners, attendants, drivers, and lab assistants. Some of these staff members may not receive institutional email addresses, while others may have limited digital literacy or may not be comfortable completing an email-led registration flow. As a result, a part of the university workforce becomes underrepresented or entirely invisible in the digital directory.

This creates both an inclusiveness problem and a practical communication gap. Support staff often play vital operational roles within departments and offices, and their absence from the system weakens the completeness of the directory. In a campus service intended to improve institutional communication, excluding one segment of the workforce reduces the value of the platform for everyone else as well.

Recommendations

An alternative authentication method, such as OTP verification through registered mobile numbers, should be introduced for specific user tiers that may not have institutional email access. In addition, authorized administrative officers should be able to create and manage read-only profiles for department support staff so that their contact information remains searchable even when they do not actively use the app themselves. Such measures would improve inclusivity and strengthen the completeness of the university directory.

2.2.5 Problem 05

Statement of the problem

Technical debt in visual data management is another notable issue in the PoC-RUET app. Significant difficulties arise when older uploaded profile photos must be managed, moderated, and verified manually over time.

Summary of findings

The existing process of manually checking user-uploaded photos to ensure that they appropriately match the employee or student creates an administrative bottleneck that becomes increasingly difficult to sustain as the system grows.

Details of findings

As the directory expands, manual review of legacy image data becomes time consuming and inefficient. Older profile images may be low in resolution, poorly cropped, outdated, visually inconsistent, or unsuitable for use in an official university directory. Without an efficient validation mechanism, the system accumulates visual inconsistency and moderation backlog, which increases the burden on the administrators responsible for maintaining the app.

This issue affects more than appearance alone. Profile photos contribute to recognition, trust, and professionalism, especially in a university setting where users may rely on images to identify faculty members, officers, or staff members correctly. If image quality varies widely or if verification is delayed, the app may appear less credible and less professionally maintained. Over time, this can reduce confidence in the quality of the platform as a whole.

Recommendations

Automated image validation should be introduced during the upload process. For example, the system may check whether a face is present, whether the image meets a minimum resolution threshold, and whether the photo follows basic profile standards. To reduce the accumulated technical debt, the university may also launch a one-time photo update campaign requiring users to re-upload compliant images. Final approval can then be delegated to departmental authorities rather than centralized entirely at the ICT Cell, thereby reducing bottlenecks and improving scalability.

2.2.6 Problem 06

Statement of the problem

A disconnect between digital contact locations and physical campus navigation is also evident in the current app design. The problem occurs when users are able to identify a person digitally but still struggle to locate the corresponding office or department physically on campus.

Summary of findings

Users may see a text-based location such as “Building A, Floor 3” or a room number, but they are not given enough spatial context to understand where that place is in relation to the RUET campus layout.

Details of findings

A plain room number or building label may be understandable to regular members of a department, but it is often insufficient for freshers, visiting guests, parents, or individuals from other departments. These users may know whom they need to meet yet still waste time trying to identify the correct building, entrance, floor, or route. In such cases, the directory successfully delivers contact information but fails to support the user's full task of actually reaching the intended person.

This gap limits the app's effectiveness as an end-to-end support tool. A modern campus directory should ideally bridge digital discovery and physical navigation, especially when one of the main goals is to make university services more accessible. Without location visualization or contextual guidance, the user experience remains incomplete.

Recommendations

Contact profiles should be integrated with a digital campus map. When a user taps on an office location, the app should open an interactive map that shows the exact building position and provides simple walking guidance where appropriate. Even a basic map-based visual cue would make the app much more helpful for new users and visitors.

2.2.7 Problem 07

Statement of the problem

Communication bottlenecks arise because the app does not yet provide integrated messaging or calling features. Users can identify the correct contact, but they must still leave the app to initiate actual communication.

Summary of findings

Users may successfully find a phone number or contact profile, yet they must manually copy the number, memorize it temporarily, or switch to an external calling or messaging application to proceed.

Details of findings

This limitation introduces unnecessary friction into an otherwise useful workflow. A directory app should not only help users discover contact details but should also streamline the next step of reaching the person. When users are forced to move between apps repeatedly, the process becomes

slower and more error-prone. This is particularly inconvenient in urgent situations or when users are attempting to contact busy faculty members and officers quickly.

In addition, regular voice calls may not always be the most suitable method of communication. Some users may prefer messaging first, while others may benefit from internal communication options that are more appropriate for institutional use. If the app functions strictly as a read-only directory, then an important opportunity is lost to make communication more efficient and user-centered.

Recommendations

The app should include direct communication links where feasible. One-tap actions could allow users to place a phone call, open a messaging channel such as WhatsApp, or connect to an internal RUET IP phone or VoIP service if such infrastructure is available. These features would transform the app from a passive information repository into a more active communication support tool.

2.2.8 Problem 08

Statement of the problem

The app is currently underutilized as a proactive campus communication tool, which contributes to low daily engagement. At present, the platform mainly acts as a passive lookup service that users open only when they need a specific contact detail.

Summary of findings

The existing usage pattern suggests that users engage with the app only on an occasional basis because it offers limited value beyond contact search. As a result, the platform is not yet functioning as a regular part of campus digital life.

Details of findings

The ICT Cell has already established a potentially valuable digital channel by deploying the app to members of the university community. However, the app is not yet being used to its full strategic potential. Universities often need a fast way to communicate urgent notices, changes in schedule, weather-related closures, emergency messages, or deadline reminders. If the PoC-RUET app remains only a static directory, then it misses the opportunity to become a more central and frequently used platform.

Low engagement can also affect long-term sustainability. Users are more likely to keep an app installed, updated, and trusted when it provides recurring value. If it is used only rarely, user attention declines and the perceived importance of the platform may diminish. This makes it harder for the app to establish a strong presence within the university’s digital ecosystem.

Recommendations

The app’s existing user base should be used to support targeted push notifications for urgent and relevant announcements. These may include sudden weather closures, broad class cancellations, administrative reminders, or important deadlines linked to RUET online services. By introducing carefully managed notification features, the app could improve daily relevance while also serving as a more effective institutional communication channel.

2.3 Conclusion

In conclusion, the initial feasibility study of the PoC-RUET directory application establishes an important foundation for the later stages of system analysis, design, and development. This study demonstrates that the value of the app lies not only in providing a centralized contact directory, but also in its potential to become a dependable digital support platform for the wider RUET community. However, achieving that goal requires careful attention to the key limitations identified throughout this chapter.

The analysis has revealed several significant concerns, including stale and inaccurate directory data, limited search flexibility, the absence of offline access, exclusion risks for lower-level support staff, inefficiencies in photo verification and moderation, weak linkage between digital records and physical campus navigation, communication friction caused by the lack of integrated contact actions, and the underuse of the app as a proactive communication channel. These issues affect not only technical performance, but also user trust, inclusiveness, operational usefulness, and long-term adoption.

By addressing these challenges through targeted improvements, the PoC-RUET app can be developed into a more reliable, accessible, and user-centered system. Features such as automated data synchronization, flexible search mechanisms, offline caching, inclusive authentication methods, scalable image validation, map-assisted location guidance, integrated communication tools, and meaningful notification services would significantly strengthen the platform. Thus, this chapter has not only identified the core problems of the current concept, but has also outlined practical directions for improvement that align with both institutional needs and user expectations.

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Chapter 3

Requirements Analysis and System Investigation

3.1 Introduction

To thoroughly understand the RUET PoC (Point of Contact) app system and identify its underlying architecture, requirements, and operational challenges, the most critical step is effective information gathering. This process involves using various tools known as information gathering tools, which play a vital role in system analysis.

In our case, information was collected in multiple stages:

1. Initial data was obtained from publicly available sources such as the RUET ICT Cell's official documentation, system specifications, and published technical reports. These provided insights into the system's objectives, technical architecture, and organizational context within RUET.
2. Direct data collection was carried out through structured interviews and questionnaires with key officials and technical personnel from the RUET ICT Cell. This approach allowed us to gather in-depth knowledge about the system's day-to-day operations, internal workflows, technical infrastructure, current capabilities, and future requirements.
3. On-site observation and system documentation review provided practical insights into actual system usage, deployment environment, and real-world operational challenges.

By combining both documentary resources and primary data collection techniques, we achieved a comprehensive view of the RUET PoC system under study.

3.2 Information Gathering Tools

To collect information from the RUET ICT Cell regarding the PoC app system, we adopted a structured and systematic approach. Our team visited the ICT Cell to directly observe the system infrastructure and operational environment. Additionally, we engaged in direct communication with the Director, Senior System Analysts, and technical personnel to gain deeper insights.

The information collection process was carried out in four distinct steps:

3.2.1 On-Site Observation

Our team observed the RUET ICT Cell operations and infrastructure, which allowed us to study how the system is deployed, maintained, and used in real-time.

3.2.2 Structured Interviews

We conducted structured interviews with key officials including the Director, Senior System Analysts, Assistant Computer Programmer, and Network Engineer to gather qualitative insights into system architecture, operational processes, current challenges, technical capabilities, and perspectives on the system's effectiveness and future requirements.

3.2.3 Questionnaires

Pre-designed questionnaires were used to collect quantitative and standardized information from the technical and administrative staff, enabling us to compare responses, identify patterns, and understand user satisfaction and requirements from multiple stakeholders.

This multi-step approach ensured that the information gathered was both comprehensive and reliable, supporting our understanding of the system from multiple perspectives.

3.3 Review of Literature, Procedure and System Documentation

Our team gathered essential information from the RUET ICT Cell's official documentation, system specifications, and technical reports, which provided a clearer understanding of the organization's background, technical structure, and system operations.

3.3.1 RUET ICT Cell Background and Organization

The RUET ICT Cell was established to provide comprehensive information technology infrastructure and digital service delivery across Rajshahi University of Engineering & Technology. The cell operates under the administrative structure of RUET and is responsible for managing institutional digital systems, supporting academic operations, and enabling inter-institutional communication.

3.3.2 Overview of RUET PoC System

The RUET PoC (Point of Contact) app is a centralized directory and contact management system designed to consolidate staff and faculty information across the university. The system serves as a baseline automation initiative, providing a single point of access for institutional contact information.

System Architecture Overview

1. **Backend Infrastructure:** Core PHP application server with MySQL relational database, deployed on BdRen C-panel managed hosting server.
2. **Database Design:** Single centralized SQL database handling all services, ensuring data consistency and eliminating historical ambiguities in employee identification and email assignment.
3. **Frontend Interfaces:** Android mobile application for end-user access, web portal at poc.ruet.ac.bd for administrative functions, and JavaScript for client-side interactivity.
4. **User Base:** Primary focus on Teachers (faculty members) and Officers (administrative staff), with future expansion planned to include Students and low-level support staff.
5. **Data Scope:** Employee directory information including name, designation, department, employee ID, contact details (mobile, IP phone, office mail, personal mail), office location hierarchy, joining date, and 300x300 photograph.

Collected System Information

Technical Stack:

1. Current Version: Core PHP, MySQL, JavaScript
2. Planned Version 2: Laravel framework, MySQL, jQuery, Tailwind CSS
3. Deployment: BdRen C-panel hosting server

4. Mobile Platform: Android primary, iOS unreleased
5. Online Dependency: Currently requires full online connectivity

Organizational Team Structure:

1. Director: Dr. Md. Samiul Habib (Overall ICT supervision and strategic planning)
2. Senior System Analysts: S Mustafa Kamal, Engr. Shihan Arafin (System analysis and technical architecture)
3. Assistant Computer Programmer: Engr. Md. Julfikar Islam (Backend development)
4. Assistant Network Engineer: Md Ashif Al Amin (Infrastructure and hosting)
5. Lab Assistant: Md. Shahnewaz Uddin (Technical support and testing)

3.4 On-Site Observation

During our visit to the RUET ICT Cell, we were genuinely impressed by the overall technical environment and professional approach to system development. Upon arrival, we first approached the administrative office and introduced ourselves, explaining the purpose of our visit related to academic coursework on system analysis.

We were directed to meet with the Senior System Analysts and technical team members. They warmly welcomed us and expressed their willingness to support our study of the PoC system.

3.4.1 Workplace Environment and Operations

While observing the ICT Cell operations, we observed the following:

1. The technical team appeared highly engaged in system maintenance and development activities, demonstrating strong collaboration and technical expertise.
2. The server infrastructure was well-maintained with proper documentation and access control measures in place to ensure data security and system reliability.
3. The team demonstrated coordinated effort in managing both current system operations and planning for Version 2 upgrades, showing both reactive maintenance and proactive improvement initiatives.

4. The working environment was professional and collaborative, with clear communication channels between administrators, developers, network engineers, and support staff.
5. Documentation practices appeared comprehensive, with technical specifications, database schemas, and system procedures well-organized and accessible to authorized personnel.

3.4.2 System Infrastructure Assessment

1. Server infrastructure was observed to be stable and properly configured on BdRen C-panel hosting platform.
2. Database systems (MySQL) were observed to maintain consistent data integrity with proper backup and access control procedures.
3. Administrative interfaces and web portal (poc.ruet.ac.bd) were functional and provided comprehensive system management capabilities.
4. Mobile app deployment was observed to be actively used by designated staff members for directory access and contact information retrieval.

3.5 Structured Interviews

When we explained our academic research purpose to the team, the Senior System Analysts appointed Engr. Shihan Arafin to conduct a structured technical interview with us, providing detailed insights into the system's architecture, operations, and requirements.

3.5.1 Interview with Senior System Analyst

System Architecture and Technical Foundation

Q1: Can you provide us with a comprehensive overview of the RUET PoC app's technical architecture and system design? The RUET PoC app is built on a centralized, relational database architecture using MySQL with a Core PHP backend server. The system follows a three-tier architecture:

1. **Presentation Layer:** Android mobile application for end-users and web-based administrative portal at poc.ruet.ac.bd.
2. **Application Layer:** Core PHP backend processing user requests, managing authentication, and handling business logic.

3. **Data Layer:** Centralized MySQL database storing all staff information, categorized into Teacher, Officer, and Staff modules.

The entire system is deployed on BdRen C-panel managed hosting infrastructure, ensuring reliable uptime and professional server management.

Q2: What data elements are currently stored in the RUET PoC system? The system maintains the following comprehensive data fields for each staff member:

1. Name and Employee ID (unique institutional identifier)
2. Designation and Department assignment
3. First Joining Date (employment start date)
4. Contact Information: Mobile Number, IP Phone, Office Mail, Personal Mail
5. Office Location: Building, Floor, Room hierarchy for precise physical localization
6. Photograph: 300x300 pixel image for visual identification and verification
7. Module Classification: Teacher, Officer, or Staff categorization

Q3: What is the data input process and workflow for populating the directory? Data entry is performed through the administrative web interface at poc.ruet.ac.bd by authorized ICT Cell personnel. The workflow follows these steps:

1. Manual data entry of staff information through standardized web forms
2. Validation and verification of entered information for accuracy
3. Categorization of staff into appropriate modules (Teacher, Officer, Staff)
4. Storage in the centralized SQL database
5. Synchronization and automatic distribution to the mobile app
6. Users access current directory information through the Android app or web portal

The system currently operates in full online mode, requiring continuous network connectivity for all functionality.

System Components and Technology Stack

Q4: Can you describe the complete technology stack used in the RUET PoC application? The current production system uses the following technology stack:

1. **Backend Framework:** Core PHP (procedural PHP without modern framework)
2. **Database:** MySQL with relational schema design
3. **Frontend (Web):** HTML/CSS with vanilla JavaScript for client-side interactivity
4. **Frontend (Mobile):** Android native application
5. **Server:** BdRen C-panel managed hosting
6. **Version Control:** Git-based repository management

For Version 2, significant technology upgrades are planned:

1. **Backend Framework:** Migration to Laravel for improved modularity and maintainability
2. **Frontend:** jQuery for enhanced client-side interactivity and Tailwind CSS for modern UI/UX design
3. **Database:** MySQL maintained with optimized schema design
4. **Architecture:** Planning for offline-first capabilities with potential Firebase synchronization

Q5: What is the current user base and scope of the RUET PoC app? The system currently serves three primary user categories:

1. **Teachers (Faculty Members):** Academic faculty and teaching staff with complete contact and location information.
2. **Officers (Administrative Staff):** Administrative and support officers managing various university functions.
3. **System Administrators:** ICT Cell personnel managing data entry, user access, and system maintenance.

User categories currently **not included** in the system:

1. Students (excluded from current scope, planned for future expansion)
2. Third and fourth class employees (low-level manual workers) - challenging to integrate due to employment structure and contact information inconsistencies

Operational Challenges and System Limitations

Q6: What are the primary technical challenges currently faced by the RUET PoC system? The system faces several significant challenges:

1. **Inclusion Gap:** The current system structure effectively serves Teachers and Officers but struggles to cleanly incorporate low-level employees (3rd/4th class workers) due to irregular employment patterns and inconsistent contact information.
2. **Online Dependency:** The system requires continuous network connectivity with no offline caching capability. This creates fragility in areas with poor network coverage and limits accessibility.
3. **Data Quality Issues:** Legacy photographs uploaded during initial deployment lack standardized verification. Manual QA processes for photo verification create bottlenecks as the system scales.
4. **Technology Stack Limitations:** Core PHP lacks modularity and maintainability features required for modern application development. The current stack makes it difficult to add new features or scale the system.

Q7: How are data accuracy and currency currently maintained in the system? Data accuracy is maintained through the following mechanisms:

1. Periodic manual verification by ICT Cell administrators
2. Direct data entry by authorized personnel with access controls
3. Email notifications to staff members requesting data updates when organizational changes occur
4. Manual review process before data goes live in the directory

However, challenges remain:

1. Delays in reflecting organizational changes from the source systems
2. Inconsistencies in photograph verification and standards
3. Limited automated validation of contact information accuracy
4. Manual processes become inefficient as the user base grows

Future Development and Enhancement Plans

Q8: What are the planned enhancements and future roadmap for the RUET PoC system? The RUET PoC system has a structured development roadmap:

Short-term Improvements (Version 1.x):

1. Add Search functionality with advanced filtering capabilities
2. Implement Add to Favorites feature for quick access to frequently accessed contacts
3. Improve user interface design and user experience
4. Enhance error handling and system reliability

Medium-term Upgrades (Version 2):

1. Complete technology stack migration to Laravel, jQuery, and Tailwind CSS
2. Implementation of offline-first architecture with local data caching
3. Firebase or alternative cloud synchronization for hybrid online/offline functionality
4. Database schema optimization and improved indexing for performance

Long-term Expansion:

1. Integration of additional institutional modules (Guest House Booking System, Postgraduate Application System)
2. Facial recognition implementation for automated photo verification
3. Student directory integration and services
4. Mobile app expansion to iOS with feature parity to Android version
5. Enhanced search and recommendation features using machine learning

Q9: How will the system address the challenge of including low-level employees? Addressing the inclusion gap requires multiple approaches:

1. **Simplified Registration Flow:** Development of streamlined data entry process for employees without regular office assignments.

2. **Phone-Based Verification:** Implementation of alternative verification methods beyond email for employees with limited institutional email usage.
3. **Flexible Data Schema:** Modifications to system to accommodate variable employment arrangements and contact information patterns.
4. **Mobile-First Approach:** Enhanced mobile app capabilities for self-service profile management and updates.

3.6 Questionnaire

To better understand system requirements, technical priorities, and satisfaction levels among technical staff, our team developed a questionnaire focusing on various aspects of system functionality, technical adequacy, and development priorities. The survey was administered to 8 technical and administrative personnel using confidential forms, each with multiple-choice options to ensure anonymity and encourage honest feedback.

Below is a question-wise breakdown with response options and summarized findings.

3.6.1 Q1: How satisfied are you with the current RUET PoC app's technical architecture?

Options:

- a. Not Satisfied
- b. Less Satisfied
- c. Maybe Satisfied
- d. Satisfied
- e. Very Satisfied

Findings:

1. 37.5% selected d. Satisfied
2. 37.5% selected c. Maybe Satisfied
3. 25% selected b. Less Satisfied

Interpretation: While the current system is functional, there is recognition that the architecture has limitations. Satisfaction levels indicate readiness for technology stack upgrades to modern frameworks.

3.6.2 Q2: How well does the current system meet the information storage and retrieval requirements?

Options:

- a. Very Poorly
- b. Poorly
- c. Adequately
- d. Well
- e. Very Well

Findings:

- 1. 50% selected d. Well
- 2. 37.5% selected c. Adequately
- 3. 12.5% selected b. Poorly

Interpretation: The system successfully stores and retrieves directory information, meeting basic functional requirements. However, some gaps exist in advanced retrieval functionality.

3.6.3 Q3: What is your priority for the next system enhancement?

Options:

- a. Offline-first functionality
- b. Technology stack upgrade (Laravel/Tailwind)
- c. Search and filter improvements
- d. Enhanced user interface
- e. System security enhancements

Findings:

1. 37.5% selected b. Technology stack upgrade
2. 25% selected a. Offline-first functionality
3. 25% selected c. Search and filter improvements
4. 12.5% selected d. Enhanced user interface

Interpretation: Technical staff prioritizes modernization of the technology stack, recognizing the importance of using contemporary frameworks for maintainability and future scalability. Offline functionality and improved search are also significant priorities.

3.6.4 Q4: How adequately does the current system handle the inclusion of diverse employee categories?

Options:

- a. Very Inadequately
- b. Inadequately
- c. Somewhat Adequately
- d. Adequately
- e. Very Adequately

Findings:

1. 50% selected b. Inadequately
2. 37.5% selected c. Somewhat Adequately
3. 12.5% selected d. Adequately

Interpretation: There is consensus that the system needs improvement in handling diverse employee categories, particularly low-level staff. This confirms the identified inclusion gap challenge.

3.6.5 Q5: Are you satisfied with the current MySQL database design and data structure?

Options:

- a. Not Satisfied
- b. Less Satisfied
- c. Maybe Satisfied
- d. Satisfied
- e. Very Satisfied

Findings:

- 1. 50% selected d. Satisfied
- 2. 37.5% selected c. Maybe Satisfied
- 3. 12.5% selected a. Not Satisfied

Interpretation: The relational database design effectively supports current requirements. However, some recognition exists that future scaling and feature additions may require schema optimization.

3.6.6 Q6: How well does the current system handle data security and confidentiality?

Options:

- a. Very Poorly
- b. Poorly
- c. Adequately
- d. Well
- e. Very Well

Findings:

1. 62.5% selected d. Well
2. 25% selected c. Adequately
3. 12.5% selected b. Poorly

Interpretation: Data security is generally well-maintained through current practices. However, some concerns exist regarding potential security enhancements as the system expands.

3.6.7 Q7: What is your overall confidence in the system's ability to scale to future requirements?

Options:

- a. Very Low
- b. Low
- c. Moderate
- d. High
- e. Very High

Findings:

1. 37.5% selected c. Moderate
2. 37.5% selected d. High
3. 25% selected b. Low

Interpretation: While the system has demonstrated foundational capability, confidence in long-term scalability is moderate. The planned Version 2 upgrade is critical to improving scalability and maintainability for future growth.

3.7 Conclusion

This comprehensive investigation of the RUET PoC app system, conducted through systematic documentation review, on-site observation, structured interviews, and staff questionnaires, has yielded significant insights into the system's current state, capabilities, and future requirements.

3.7.1 Summary of Findings

The RUET PoC app represents a successful initial automation effort at RUET, establishing a centralized, authoritative source for staff directory information. The single-database SQL design ensures data consistency, eliminates historical ambiguities in employee identification and email assignment, and provides reliable operational functionality for Teachers and Officers.

3.7.2 System Strengths

1. Stable and reliable centralized database architecture
2. Clear organizational team structure with defined responsibilities
3. Well-maintained server infrastructure on professional hosting platform
4. Effective data storage and retrieval mechanisms for current user base
5. Professional approach to system maintenance and development
6. Clear development roadmap and modernization strategy

3.7.3 Key System Characteristics

1. **Technology Stack:** Core PHP (current), planned migration to Laravel/Tailwind (Version 2)
2. **Data Architecture:** Centralized MySQL database with relational design
3. **User Base:** Teachers and Officers primarily, with planned expansion to Students
4. **Operational Mode:** Full online dependency in current version, offline capability planned for Version 2
5. **Data Scope:** Comprehensive contact and location information for all categorized staff members

3.7.4 Key Requirements Identified

1. Technology stack modernization to Laravel/Tailwind/jQuery framework stack
2. Implementation of offline-first architecture with Firebase synchronization
3. Development of mechanisms for low-level employee inclusion
4. Enhancement of search and filtering capabilities

5. Implementation of automated photo verification using facial recognition
6. Integration of future institutional modules and services
7. User interface improvements and enhanced user experience

3.7.5 Strategic Recommendations

The investigation indicates that the RUET PoC system is well-positioned for continued development and enhancement. Priority should be given to:

1. Accelerating Version 2 technology stack migration to improve maintainability and scalability
2. Implementing offline-first functionality to enhance system reliability in diverse network environments
3. Developing inclusive mechanisms for low-level employee integration
4. Establishing automated data quality processes to reduce manual verification overhead
5. Planning for modular architecture to support future institutional module integration

Through systematic implementation of these requirements and recommendations, the RUET PoC app can evolve into a robust, scalable platform serving the complete institutional communication and directory needs of RUET.